

Topological Strain Memory on the $6N$ Skeleton: the Effective Second-Order Markov Dynamics of Prime Wing Transitions*

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada

ruqing@hotmail.com

June 8, 2026

Abstract

Every prime > 3 sits on a wing of the $6N$ lattice—right ($6N+1$) or left ($6N-1$)—so the prime sequence is a binary string of wing symbols. Lemke Oliver and Soundararajan showed that consecutive primes avoid repeating their residue class; here we ask a sharper question of the wing string: *how deep is its memory?* Using all 26,355,865 primes $3 < p \leq 5 \times 10^8$, we test the Markov order of the chain. It is decisively *not* first-order: the probability of staying on a wing depends on the previous two primes, not one ($P(R | RR) = 0.4252$ versus $P(R | LR) = 0.4580$, a gap of 0.033 at 168σ). The cleanest probe is the switch (step) sequence, which for a symmetric first-order chain would be independent noise: instead it carries an autocorrelation of -0.0328 at lag one (a switch is followed by a switch less often than chance—we name this a topological strain memory), a weak $+0.0046$ echo at lag two, and *nothing* measurable beyond. The incremental memory contributed by each further prime halves per order ($0.033, 0.016, 0.009, \dots$), so the chain is *effectively*—not exactly—second-order: a dominant nearest-neighbour memory with a rapidly decaying tail. We then show the wing memory and the well-known anti-correlation of consecutive prime gaps are *the same effect*: the switch indicator is identically the event “gap $\not\equiv 0 \pmod{6}$,” and the step lag-one autocorrelation (-0.0328) equals the normalised-gap lag-one correlation (-0.0332). All signals fade slowly with scale, as the Hardy–Littlewood framework requires. This is a measurement, not a theorem: the structure lives inside the Hardy–Littlewood r -tuple heuristic and the Oliver–Soundararajan bias, and what we add is the explicit memory-depth characterisation and the gap identification.

1 Introduction

A prime $p > 3$ is $\equiv 1$ or $\equiv 5 \pmod{6}$; call these the right wing ($6N+1$) and the left wing ($6N-1$). Reading the primes in order turns them into a string $\dots \text{RLRRLR} \dots$, and the naive Cramér model says it should be a fair coin: each wing equally likely, transitions independent. It is not. Lemke Oliver and Soundararajan [2] found that consecutive primes are biased *against* repeating their class modulo q ; for $q = 3$ —which is exactly our wing label, since $6N+1 \equiv 1$ and $6N-1 \equiv 2 \pmod{3}$ —this makes same-wing transitions rarer than switches.

The bias itself is by now expected. Our question is one level deeper and, to our knowledge, not usually posed directly: *what is the memory depth of the wing string?* It is sharpest to set three

*Code and measured data: <https://github.com/Ruqing1963/6N-wing-transitions>

pattern	count	$P(\text{next} = R)$	pattern	count	$P(\text{next} = R)$
RR	5,843,102	0.4252	LR	7,334,397	0.4580
RL	7,334,397	0.5420	LL	5,843,967	0.5748

Table 1: Second-order conditional probabilities. Holding the current wing fixed, the next wing still depends on the previous one ($P(R | RR) \neq P(R | LR)$), so the chain is not first-order Markov.

targets against which to read the answer. The *zeroth* target is the Cramér model: the wings are a fair coin, switches unbiased and independent. The *first* target is the naive reading of the Oliver–Soundararajan bias: a switch bias exists, but lives entirely in the current step, so that the chance of switching does not depend on how the current wing was reached— $P(\text{switch} | \text{previous step})$ is constant. The *third* possibility, which the data select, is that the chain is genuinely higher-order: the switch probability depends on the previous step, and the dependence decays across further steps rather than terminating at once. The first target is broken decisively; the third is what we measure, with the memory dominated by the second order and a geometrically decaying—not abruptly truncated—tail. This is a sharp, finite, measurable question with an answer we did not know in advance, and we answer it with the full prime census to 5×10^8 .

We are explicit from the outset about what is and is not new. The existence of higher-order structure is contained in the Hardy–Littlewood r -tuple heuristic [1], which predicts the frequency of every finite pattern of consecutive primes, and the Oliver–Soundararajan analysis [2] is built on it. We prove no theorem and propose no new mechanism. What this paper contributes is the explicit characterisation of the memory—its order, its sign, its decay rate—in the geometric language of the $6N$ wings, and the identification of that memory with the anti-correlation of consecutive prime gaps.

2 Setup: the wing string and its switch sequence

Let $w_i \in \{R, L\}$ be the wing of the i -th prime > 3 . The two wings occur almost equally often (to 5×10^8 , 13,177,500 right against 13,178,365 left; the residual Chebyshev bias is below 10^{-4}), so the chain is very nearly symmetric under $R \leftrightarrow L$.

This symmetry gives a clean test of Markov order. Define the *switch* (step) sequence $\sigma_i = \mathbf{1}[w_{i+1} \neq w_i]$. For a *symmetric* first-order Markov chain the switch probability is the same in both states, so σ_i would be an independent Bernoulli sequence with *zero* autocorrelation at every lag. Any measured autocorrelation of σ is therefore direct evidence that the wing chain is *not* first-order. We enumerate the primes by a sieve to $N = 5 \times 10^8$ and form w , σ , and the prime gaps $d_i = p_{i+1} - p_i$ directly; no modelling enters.

3 The chain is not first-order

The marginal transition is $P(\text{same wing}) = 0.441$, $P(\text{switch}) = 0.559$: already the 6% Oliver–Soundararajan deficit against the fair-coin 0.5. Conditioning on one more prime breaks the first-order hypothesis outright (Table 1): with the current wing held at R, the next wing still depends on the one before,

$$P(R | RR) = 0.4252, \quad P(R | LR) = 0.4580,$$

a difference of 0.0328. With 5.8×10^6 samples in each cell the standard error is $\sim 2 \times 10^{-4}$, so the gap is significant at 168σ . A first-order chain requires these to be equal; they are not.

4 The memory depth

How far back does the dependence reach? The switch-sequence autocorrelation (Fig. 1(A)) answers cleanly, because under the first-order null it should vanish at all lags:

$$\rho_1 = -0.0328 \ (168\sigma), \quad \rho_2 = +0.0046 \ (24\sigma), \quad \rho_3 = -1 \times 10^{-5} \ (0\sigma),$$

with $\rho_{\geq 3}$ indistinguishable from zero. The memory is overwhelmingly nearest-neighbour: a switch is *less* likely to be followed by another switch ($P(\sigma=1 \mid \sigma_{\text{prev}}=1) = 0.544$ against $P(\sigma=1 \mid \sigma_{\text{prev}}=0) = 0.578$), as if a jump between wings relieves a strain that then takes a step to rebuild—a kind of arithmetic inertia, the sequence resisting two abrupt moves in a row. We use the image only as a name for the measured sign of ρ_1 ; it describes an autocorrelation, not a force. The sign—anti-persistence of switches—is the dynamical face of the Oliver–Soundararajan deficit.

“Effectively second-order” must be stated carefully, and the full conditional table makes the qualification honest. The incremental memory contributed by the oldest prime in a pattern—the average of $|P(R \mid R \cdots) - P(R \mid L \cdots)|$ over the intervening symbols—does not cut off abruptly; it *halves* with each order (Fig. 1(B)):

$$0.0328 \ (m=2) \rightarrow 0.0159 \ (m=3) \rightarrow 0.0092 \ (m=4),$$

ratios 0.48 and 0.58, descending toward the finite-sample floor by $m \approx 5$. So the chain is not exactly a second-order Markov chain; it is one with a dominant second-order term and a geometrically decaying tail of weaker higher-order corrections. “Effective second-order” names this hierarchy, not a hard truncation. The switch autocorrelation, which already collapses by lag three, and the slower conditional decay are two views of the same short-ranged, rapidly fading memory.

5 One effect, two projections: wings and gaps

The wing memory is not separate from the known anti-correlation of consecutive prime gaps—it is the same statistic. A switch occurs exactly when the gap is not a multiple of 6: to 5×10^8 the identity $\sigma_i = \mathbf{1}[6 \nmid d_i]$ holds for every prime (agreement 1.0000), because crossing wings changes the residue mod 3. Measuring both,

$$\rho_1(\sigma) = -0.0328, \quad \text{corr}(g_i, g_{i+1}) = -0.0332 \quad (g_i = d_i / \log p_i),$$

which agree to the third decimal. The conditional gap curve (Fig. 2(A)) shows the mechanism in gap language: after a small gap the next gap runs above average, after a large gap below—primes resist clustering, and a wing switch is the coarse, mod-6 shadow of that resistance. We state this as an identification, not a metaphor: the second-order wing memory and the consecutive-gap anti-correlation are one anti-clustering tendency seen through two resolutions.

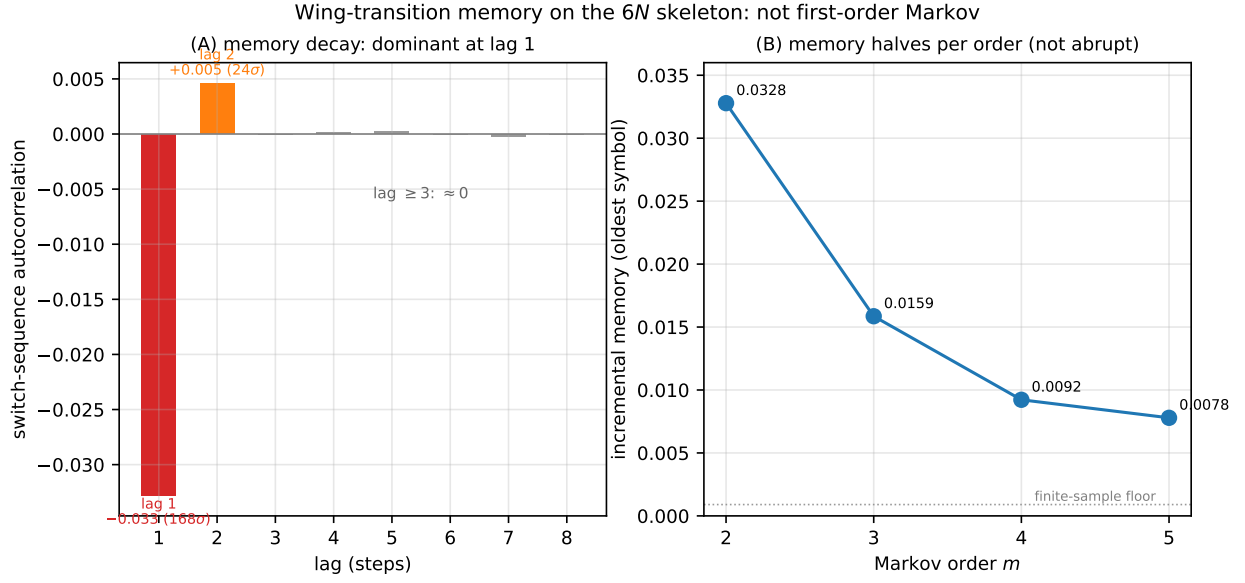


Figure 1: (A) Autocorrelation of the switch sequence: under a first-order chain it would be zero everywhere. It is -0.033 at lag one (168σ), a weak $+0.005$ at lag two, and consistent with zero beyond. (B) The incremental memory contributed by each further prime halves per order, descending toward the finite-sample floor—a decaying tail, not an abrupt cutoff.

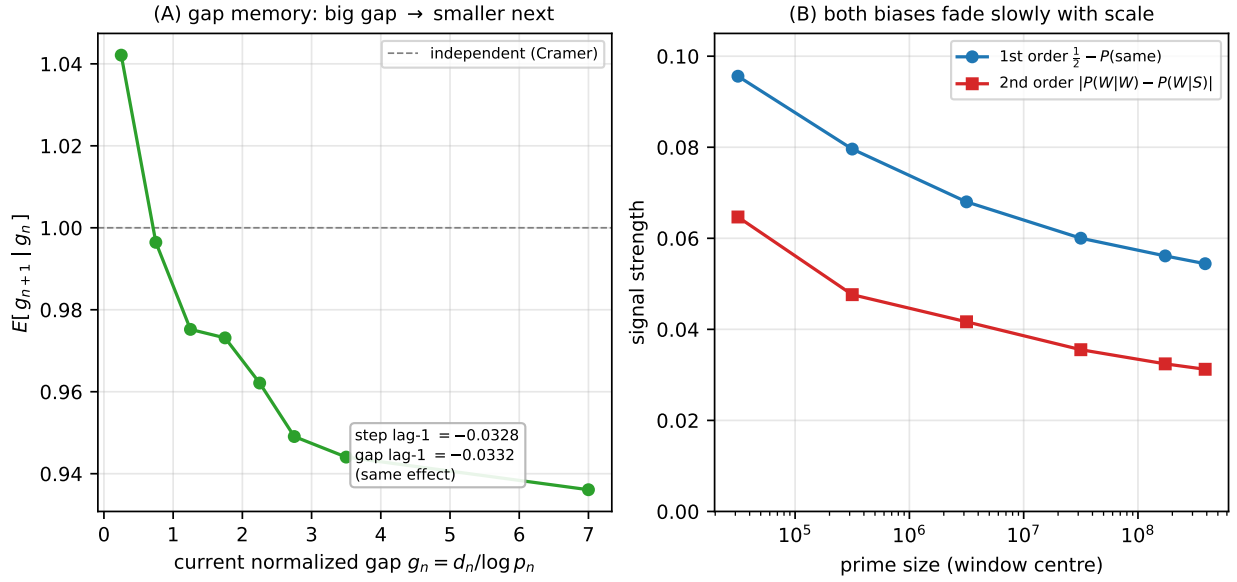


Figure 2: (A) Gap memory: the expected next normalised gap decreases with the current one (independence would be flat at 1). The wing switch lag-one autocorrelation and the normalised-gap lag-one correlation coincide (-0.0328 vs -0.0332): one anti-clustering effect, two projections. (B) Both the first-order ($\frac{1}{2} - P(\text{same})$) and second-order ($|P(W|W) - P(W|S)|$) signals fade slowly as the primes grow, as the Hardy–Littlewood framework requires.

6 Scope and honest ledger

The wing string of the primes is not a fair coin and not a first-order Markov chain. It carries a nearest-neighbour memory—a switch suppresses the next switch by 0.033—with a weak second echo and a geometrically decaying higher-order tail, so it is effectively, not exactly, second-order. That memory is the same object as the anti-correlation of consecutive gaps, the switch being identically the event that the gap is not divisible by 6. Every signal fades slowly toward the fair-coin limit as the primes grow (Fig. 2(B)): $P(\text{same})$ rises from 0.404 below 10^5 to 0.446 near 5×10^8 , and the second-order term shrinks in step.

We are explicit about the boundary. The anti-repetition bias is Lemke Oliver and Soundararajan’s [2]; the heuristic that predicts every consecutive-pattern frequency, and hence all of the structure measured here, is Hardy and Littlewood’s [1]; the slow fade with scale is the expected behaviour of these conjectural densities. We prove nothing and propose no mechanism beyond these. What this paper adds is the explicit memory-depth characterisation of the wing chain—its order, the sign and magnitude of each term, and the geometric decay—together with the identification of that memory with the consecutive-gap anti-correlation, all in the $6N$ geometric language and measured against the full prime census to 5×10^8 . The words “strain” and “anti-clustering” name measured correlations; they are not claims about a physical force. Everything here equals a direct count over the primes and takes the Hardy–Littlewood heuristic as input.

Data and code availability. The sieve to 5×10^8 , the wing and switch sequences, the order-1 to order-4 transition tables, the autocorrelation and memory-depth measurements, the gap identification, and the figure scripts are openly available at <https://github.com/Ruqing1963/6N-wing-transitions>.

References

- [1] G. H. Hardy, J. E. Littlewood, *Some problems of ‘Partitio Numerorum’; III*, Acta Math. **44** (1923), 1–70.
- [2] R. J. Lemke Oliver, K. Soundararajan, *Unexpected biases in the distribution of consecutive primes*, Proc. Natl. Acad. Sci. USA **113** (2016), E4446–E4454.
- [3] R. Chen, *Twin counts in short intervals: singular-series rigidity and the density-gradient variance* (Part XXIX), Zenodo, doi:10.5281/zenodo.20597725 (2026).
- [4] R. Chen, *The Fermat-type dodge ladder: primes $n^{2^k} + 1$ on the $6N$ skeleton* (Part XXXIII), Zenodo, doi:10.5281/zenodo.20602622 (2026).
- [5] R. Chen, *Arithmetic geodynamics on the $6N$ skeleton: a review of Parts I–XIX*, Zenodo, doi:10.5281/zenodo.20585301 (2026).